

STRENGTHENING MOBILE VPN AND HOTSPOT DEFENCES

When it comes to encryption on the go, only 32 percent of mobile users leverage VPNs daily, leaving the majority exposed to eavesdroppers. Many top-tier VPN clients include a “kill-switch” that halts all network traffic if the tunnel drops, eliminating data leaks. Always favor providers that support modern protocols like WireGuard for superior speeds and cryptographic hygiene. Remember that your phone’s VPN extends to any devices tethered via its hotspot, ensuring encrypted passthrough for laptops and tablets. Enterprises can push per-app VPN profiles through MDM, enforcing secure tunnels without user intervention. Layer in DNS-level filtering – such as CleanBrowsing’s Security filter – to block phishing and malware domains before connections resolve.

On iOS, enable “Connect On Demand” so that any network automatically triggers the VPN, even on legacy hotspots. Before departure, validate your setup with an IP/DNS leak test at ipleak.net to confirm zero leaks. Rotate your VPN credentials or encryption keys every 90 days to limit the blast radius if they’re compromised. Employ split-tunneling sparingly – restrict it to non-critical traffic to avoid accidental exposure of sensitive sessions. For ultra-secure workflows, consider multi-hop or double-VPN configurations, chaining servers across jurisdictions. And steer clear of free VPN services – 85 percent are known to log or sell user data – opting instead for audited, no-log providers with transparent policies.

3. Configure auto-connect

- In the VPN app’s settings, enable Auto Reconnect and Always-On VPN (requires Android 7+).

4. Lockdown mode (optional)

- Enabling Block connections without VPN ensures no traffic leaks if the VPN drops.

5. Test through speed and leak tools

- Use apps like Speedtest and DNS leak tests to confirm encryption.

Tip: For enterprise deployments, use MDM to auto-push VPN profiles and enforce always-on settings without user intervention.

Bluetooth, NFC & peripheral attack mitigation

Bluetooth exploits – such as BlueBorne – allow attackers to execute code or snoop data on pairing-enabled devices.

1 Keep firmware updated

- Regularly install OS and firmware updates: Settings → About → System Updates.

2. Disable when unused

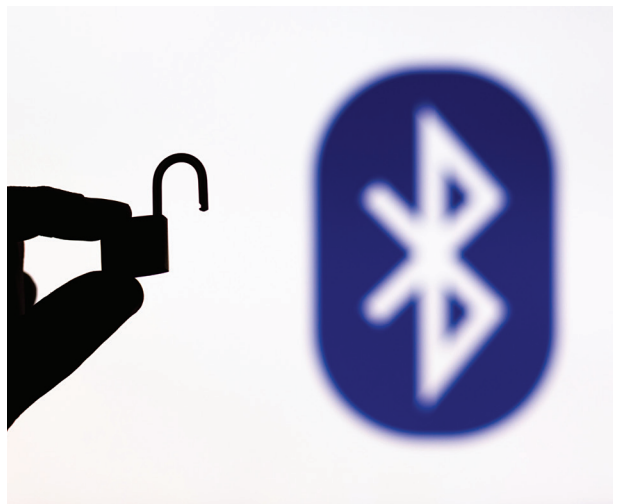
- Turn off Bluetooth in Quick Settings or Control Center; only enable during active pairing sessions.

3. Use “Non-Discoverable” mode

- Pair in discoverable mode temporarily, then switch to hidden to prevent unsolicited connections.

4. Verify pairing codes

- Confirm that the PIN on both devices matches exactly before accepting the pairing request.



NFC security measures

NFC can be abused via relay or eavesdropping attacks if left enabled continuously.

1. Enable NFC on-demand

- Settings → Connections → NFC (Android), Control Center toggle (iOS) – only turn on when needed.

2. Use secure elements for payments

- Rely on built-in hardware elements (e.g., Secure Enclave) for wallet transactions – not third-party apps.